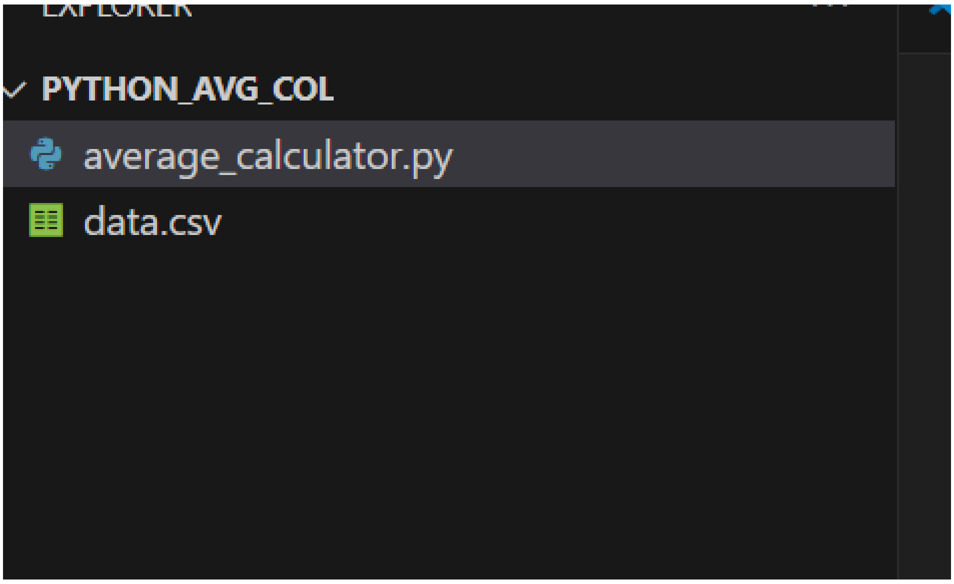
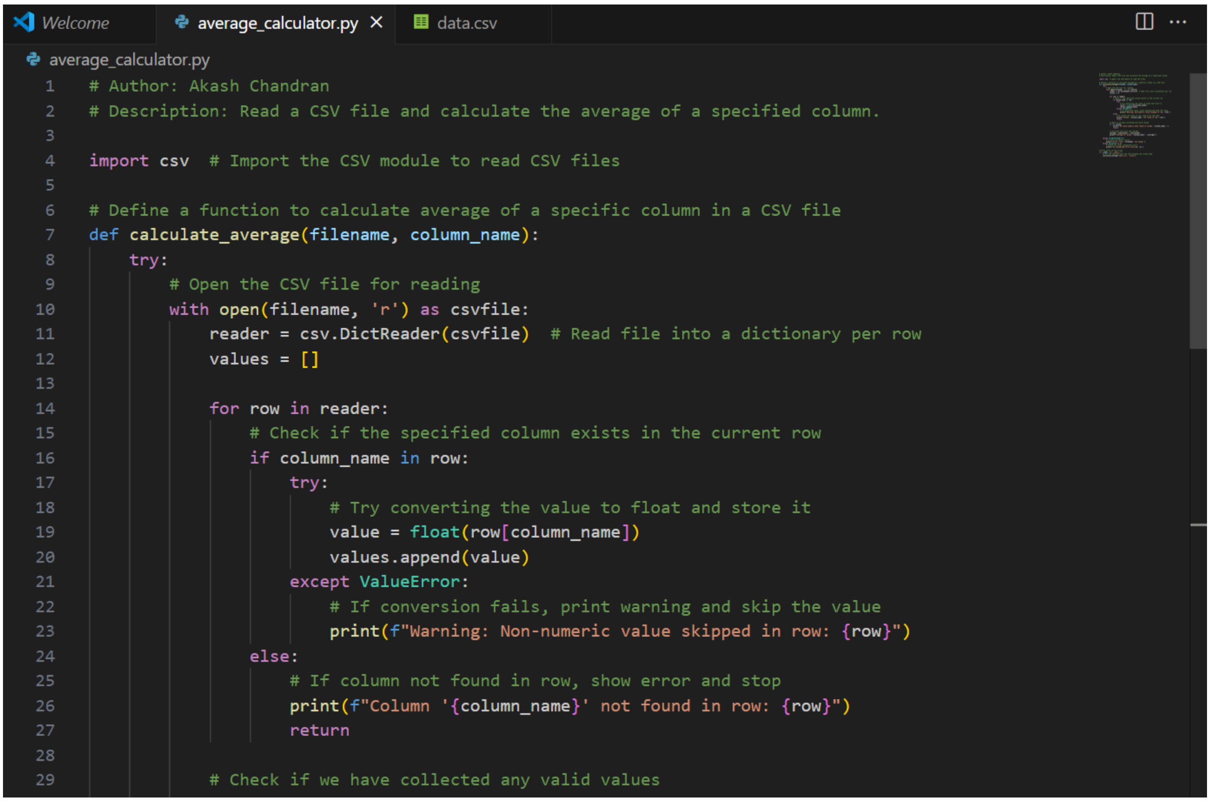
//Q2

//https://github.com/AkashChand6n/Devops-restest.git

Python script that reads a CSV file and calculates the **average value of a specified column**






```python
# Author: Akash Chandran
# Description: Read a CSV file and calculate the average of a specified column.

import csv  # Import the CSV module to read CSV files

# Define a function to calculate average of a specific column in a CSV file
def calculate_average(filename, column_name):
    try:
        # Open the CSV file for reading
        with open(filename, 'r') as csvfile:
            reader = csv.DictReader(csvfile)  # Read file into a dictionary per row
            values = []

            for row in reader:
                # Check if the specified column exists in the current row
                if column_name in row:
                    try:
                        # Try converting the value to float and store it
                        value = float(row[column_name])
                        values.append(value)
                    except ValueError:
                        # If conversion fails, print warning and skip the value
                        print(f"Warning: Non-numeric value skipped in row: {row}")
                else:
                    # If column not found in row, show error and stop
                    print(f"Column '{column_name}' not found in row: {row}")
                    return

            # Check if we have collected any valid values
```

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                # Check if we have collected any valid values
                if not values:
                    print(f"No valid numeric data found in column '{column_name}'.")
                    return

                # Calculate and print the average
                average = sum(values) / len(values)
                print(f"Average of column '{column_name}': {average}")

        except FileNotFoundError:
            # If the file doesn't exist
            print(f"Error: File '{filename}' not found.")
        except Exception as e:
            # Handle any other unexpected errors
            print(f"An unexpected error occurred: {e}")

# Entry point of the script
if __name__ == "__main__":
    # Call the function with the CSV filename and column name
    calculate_average('data.csv', 'score')
```

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sample data

```
data.csv
name,score,age
Akash,80,22
Don,90,25
vishnu,85,23
David,not_a_number,24
```

```
\Desktop\python_avg_col> python3 average_calculator.py
```

```
Warning: Non-numeric value skipped in row: {'name': 'David', 'score': 'not_a_number', 'age': '24'}
Average of column 'score': 85.0
```